

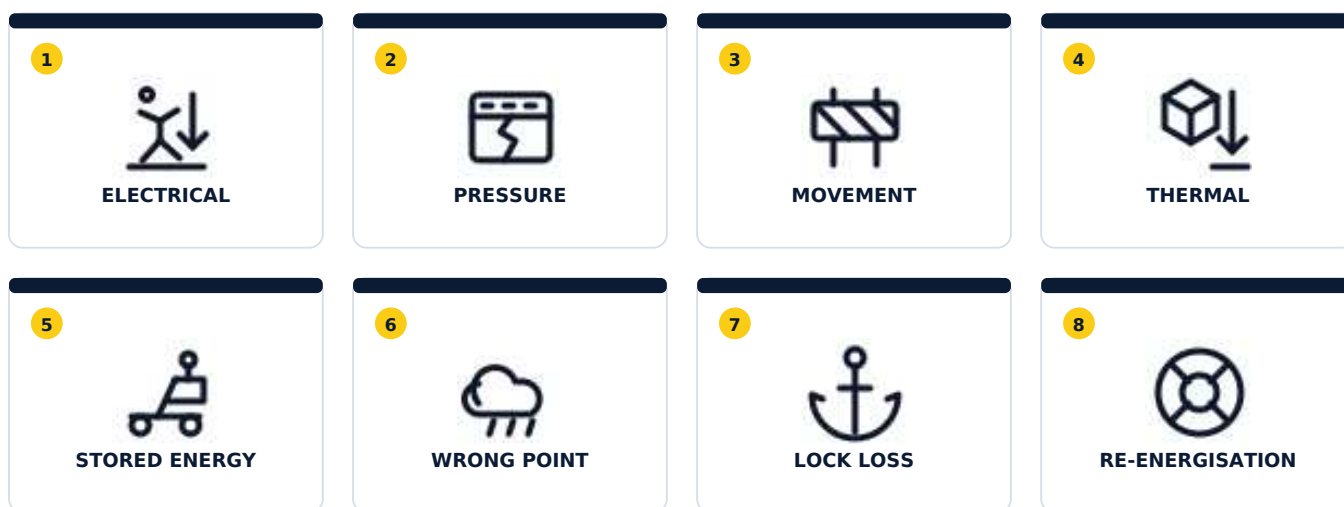
LOCKOUT TAGOUT

Prevent unexpected energisation, movement and release of hazardous energy.

PURPOSE AND SCOPE

Safely isolate electrical, mechanical, hydraulic, pneumatic, thermal, chemical and stored energy during maintenance, cleaning, adjustment and inspection work, and prevent unexpected start-up or hazardous energy release. LOTO is not limited to switching off electrical power. Every energy source must be identified, isolated at the correct point, locked, tagged and relieved of stored energy before work begins.

KEY HAZARDS



CRITICAL VALUES TO VERIFY IN THE FIELD

100%

Identification of every energy source

1 person 1 lock

Personal control principle

Zero energy

Verified condition before work



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.147 requires an energy-control procedure, authorization, personal locks and verification of zero energy.

OSHA 1910.147(c)

ENERGY CONTROL PROGRAM

Provide equipment-specific written procedures, training and periodic inspections.

OSHA 1910.147(d)

APPLICATION STEPS

Complete preparation, shutdown, isolation, lockout, stored-energy control and verification in sequence.

OSHA 1910.147(f)(3)

GROUP LOCKOUT

Each authorized employee must retain personal protection within a group-lockout arrangement.

OSHA 1910.147(f)(4)

SHIFT CHANGE

Maintain isolation continuity through an approved transfer during shift or personnel changes.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Equipment-specific LOTO procedure | ☐ Isolation-point list |
| ☐ Personal-lock record | ☐ Zero-energy test record |
| ☐ Group lock-box control | ☐ Shift-transfer record |

SUPERVISOR DISCUSSION POINTS

- Have all energy sources been identified?
- Which isolation points will be locked?
- How will stored energy be released or restrained?

PERSONAL LOCKS AND ZERO-ENERGY VERIFICATION



All sources - Personal lock - Pressure release - Test - Try-start

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. IDENTIFY ENERGY

Identify all energy sources and isolation points. Shut down equipment using the normal stop procedure.

2. ISOLATE AND LOCK

Physically isolate every hazardous energy source. Each worker must apply a personal lock and tag. Release, restrain or block all stored energy.

3. CONTROL STORED ENERGY

Test electrical isolation with a suitable test instrument. Complete a try-start test to confirm no operation. Communicate the isolation boundary to the full team.

4. VERIFY ZERO ENERGY

Apply the approved shift-change lock-transfer procedure. Check the area and personnel before removing locks.

CRITICAL WARNING



Switched-off equipment is not necessarily safe. Equipment is safe only when it is isolated, locked, tagged and verified at zero energy.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|--|
| ☐ Isolation plan ready | ☐ Energy sources identified |
| ☐ Equipment shut down | ☐ Isolation points secured |
| ☐ Personal locks applied | ☐ Tags readable |
| ☐ Stored energy released | ☐ Zero energy verified |
| ☐ Try-start completed | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			